Hemoglobin

Describe the structure, synthesis, functions and breakdown of Haemoglobin. Add a note on Haemoglobin estimation

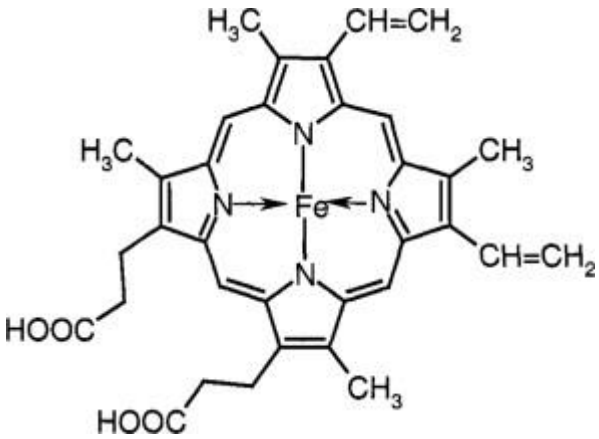
Haemoglobin is the respiratory pigment found in red blood cells, synthesized in immature erythrocytes, it gives red color to blood and is iron containing metalloprotein, consisting of heme and globin

Structure of haemoglobin:

Different types of haemoglobin molecules produced by humans and all these are tetramers composed of four globular proteins. The most predominant haemoglobin is HbA1 which contains 2α and 2β subunits. The α chain contains 141 amino acids and β chain contains 146 amino acids residues

Heme group consists of an iron(ferrous) and is held in heterocyclic ring known as porphyrin. The ring consists of 4 pyrrole cyclically arranged and are linked by methane bridge and the structure is called protoporphyrin and when it combines with iron and the resultant structure is called metalloporphyrin.

With the help of four bonds iron is attached to 4 pyrrole and with the fifth bond it is attached to histidine residue of globin and through sixth bond the iron attaches to oxygen



Synthesis of haemoglobin

Heme synthesis occurs in mitochondria

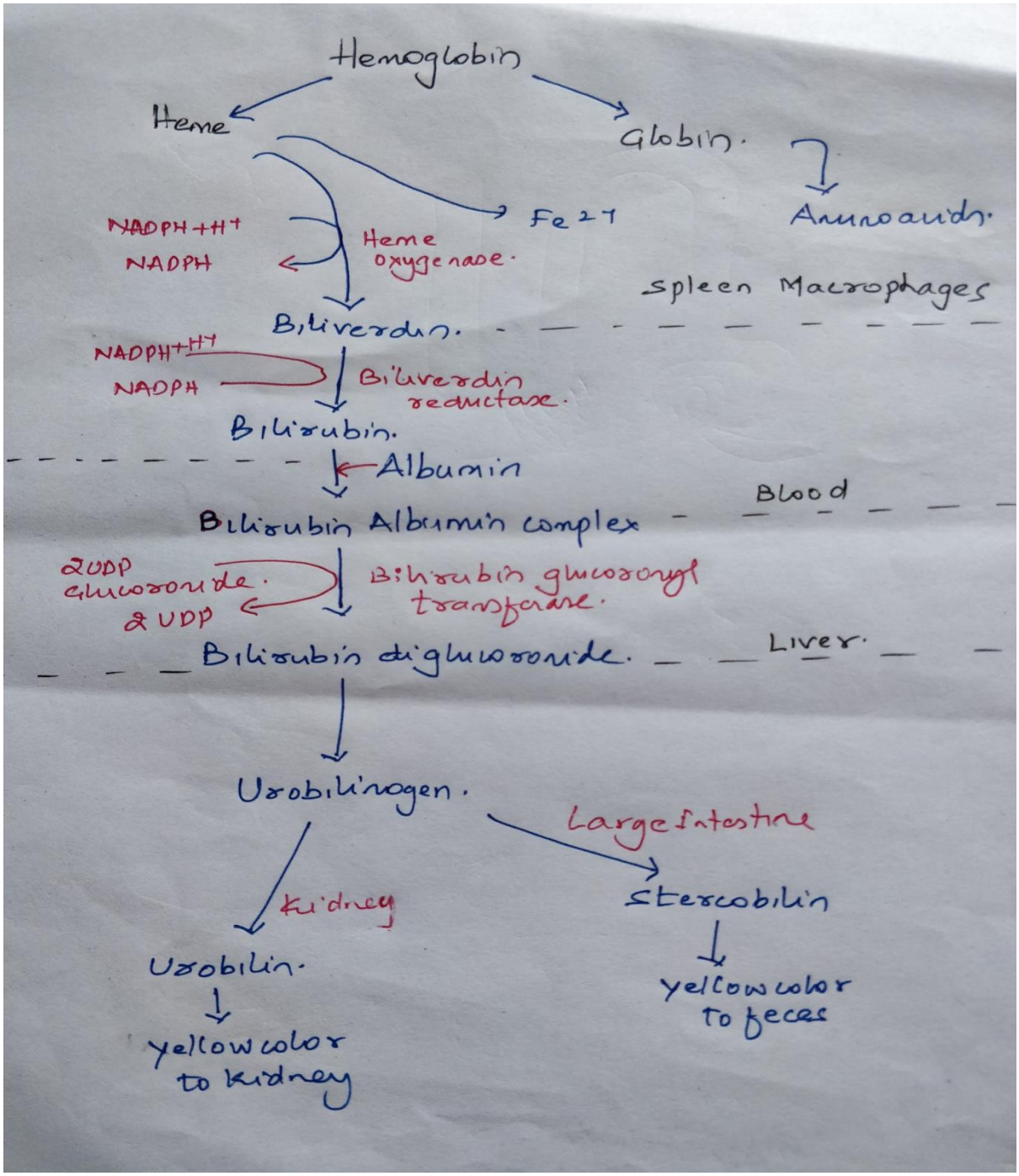
heme starts with condensation of glycine with succinyl conenzyme A to give δ-laevulenic acid and the reaction is promoted by enzyme δ amino laevulunic acid synthase and pyridoxal phosphate(Vit B6) acts as a conenzyme.

- Two molecules of δ-laevulenic acid condense to form a pyrrole called porphobilinogen
- Four molecules of porphobilinogen condense to form uroporphyrinogen III
- Uroporphyrinogen III gets converted to coproporphyrinogen
- Coproporphyrinogen then gets converted to protoporphyrin
- Protoporphyrin combines with iron(Fe2+) to form heme
- Each molecule of heme combines with globin a tetramer of 4 globin chain, each with its own heme group combine together to form haemoglobin

Breakdown of haemoglobin:

- RBC gets destroyed in spleen after 120 days and haemoglobin gets released from RBC.

- Haemoglobin split into heme and globin. Globin breakdown into amino acids and utilized by body
- The iron from heme part gets separated out and utilized for further synthesis haemoglobin
- The remaining part of heme undergoes following metabolism



Haemoglobin estimation:

Estimation of haemoglobin is done by Sahli hellige's method or Sahli;s method or acid Hematin method

The principle involved in estimation is when haemoglobin is treated with N/10 Hcl, the haemoglobin is converted to acid hematin which produce reddish brown color which is then compared with standard

Procedure:

Take N/10 Hcl upto 3g mark in haemoglobin tube

Stelize and Prick the ring finger and collect the blood in haemoglobin pipette upto 20 cumm mark

Transfer the blood immediately to sahli tube containing Hcl and mix thoroughly

Wait for 10 min during which RBC ruptures and releases haemoglobin and haemoglobin reacts with Hcl and gets converted to hematin(brown color)

After 10 min add few drops of distilled water slowly and compared with comparator

Distilled water is added in small quantity till the color in the tube matches with standard in comparator

When the color matches take the reading for hemoglobin